Recursive "What If?" Imagination in AMLD

Chess-Style Counterfactual Search Guided by MAPQUEST

Brian Tenneson

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Abstract

This note formalizes chess-style recursive counterfactual exploration in an AMLD-style verifier-gated theorem search. A sound monotone certified BANK is kept separate from speculative assumptions. MAPQUEST navigates tagged speculative branches and proposes valuable missing lemmas as proof goals; only independently checked derivations may enter BANK. The framework uses branch-addressed worlds, frozen-judgment checking, finite candidate-action windows, and the declared AMLD+0 role labels. It distinguishes consistency filtering, conditional navigation, and actual verified settlement. It makes no claim that speculative search itself discovers a proof, that the rank score is exact, or that the mechanism speeds up all inputs.

1 Framework: MAPQUEST and verifier sovereignty

Fix an effectively presented proof calculus F, authorized assumptions Γ, target well-formed formula T, independent sound certificate verifier V, and legal, fully charged action protocol. A certified state B is a finite or effectively represented set of permitted assumptions and accepted consequences with dependency evidence; writing $B \subseteq \mathrm{Th}_F(\Gamma)$ refers only to formulas whose open hypotheses have been discharged or established from Γ. The immutable target and verifier are not altered by imagination.

MAPQUEST supplies an address and traversal system for searching legal inference applications. If a rule r has arity $a(r)$, its tagged coordinate space may be represented schematically by

$$\mathcal{G} = \coprod_{r \in \mathcal{R}} (\{r\} \times (\mathbb{Q} \cap [0,1])^{a(r)}).$$

This is a representation and scheduling device, not a new inference rule or a proof of search completeness. Formula-address locations and rule-application coordinates are distinct.

2 Recursive "what if?" imagination and branch semantics

A union of hypothetical formula sets can conflate mutually exclusive branches, erase the identity of alternative assumptions, and treat an inconsistent union as one imagined world. Instead, define a *family of tagged branches*, not a union of their contents.

Definition 2.1 (Branch-preserving operator)**.** Fix an explicit node cap $N_{\max} \in \mathbb{N}_{\geq 1}$, finite candidate-window cap $K \in \mathbb{N}_{\geq 1}$, and depth $d \in \mathbb{N}$. The generator must enforce a bounded set $\Sigma_{d,N_{\max},K}$ of at most $N_{\max}$ branch addresses, each of length at most d. A charge budget b is accounted separately; its finiteness alone does not guarantee termination when operations can have zero cost. The selected set of branch addresses is finite and prefix-closed. The root has address ϵ and assumption set $H_\epsilon = \varnothing$. A child σi generated using candidate formula $L_{\sigma i}$ has

$$H_{\sigma i} = H_\sigma \cup \{L_{\sigma i}\}.$$

Define

$$\boxed{\mathcal{W}_{d,N_{\max},K}(B,T) = \{(\sigma, H_\sigma, \widetilde{B}_\sigma) : \sigma \in \Sigma_{d,N_{\max},K},\ \widetilde{B}_\sigma = B \cup H_\sigma\}.}$$

All addresses, including the root $(\epsilon, \varnothing, B)$, are retained. Formula membership in $\widetilde{B}_\sigma$ is *hypothetical*, even if a navigation algorithm can derive T from it. No element of a speculative branch may directly become a certified BANK entry.

A branch may have no admissible children and is then retained as a leaf. Branches with the same formula set remain distinct search histories when provenance or costs differ. Branches whose hypothetical assumptions are inconsistent with each other or with Γ are flagged as inconsistent and excluded from any ordinary-classical leverage estimate unless the scoring method explicitly discounts explosion. For example $H = \{p, \neg p\}$ makes T classically derivable for every T, but this is not useful evidence of $\Gamma \vdash T$. No speculative proof can enter BANK. Finiteness follows from the explicit address, depth, and candidate caps, independently of the cost model. Each generator step has a documented charged cost, including rejected branches. The explicit cap, not a mere finite numeric credit that could allow infinitely many zero-cost steps, guarantees bounded enumeration.

2.1 Chess analogy and the search boundary

A chess player asks "What if I sacrifice a piece and recover position later?" The proof-search analogue provisionally assumes an unusual lemma L, tests what it would make accessible, and then seeks an actual V-accepted proof of $\Gamma \vdash L$. An imagined winning line is no more a verified proof than a favorable hypothetical chess line is a forced win. If adversarial replies matter, a strategy requires the proper quantifier structure, not a single favorable branch.

3 BANK safety and the route back to reality

Theorem 3.1 (Counterfactual conservativity)**.** *Assume: (i) initially $B_0 \subseteq \mathrm{Th}_F(\Gamma)$ with provenance; (ii) every certified deposit of a formula φ is accepted by a verifier sound for the frozen judgment $\Gamma \vdash \varphi$, with all other undischarged assumptions rejected; (iii) speculative branches cannot write to BANK or modify V, Γ, T; and (iv) the BANK update operation contains only accepted deposits. Then every finite certified stage t satisfies*

$$B_t \subseteq \mathrm{Th}_F(\Gamma).$$

The claim holds regardless of the number, contents, or incompatibility of speculative branches; it does not assert eventual settlement.

Proof. The base case is (i). If $B_t \subseteq \mathrm{Th}_F(\Gamma)$, then by (ii)–(iv) each new accepted deposit is also in $\mathrm{Th}_F(\Gamma)$. Thus B_{t+1} is contained in the same closure. Induction completes the proof. Only finite BANK stage indices are used; arbitrary speculative branching introduces no new inference license. □

Remark 3.2 (Hypotheses really matter)**.** A proof of $\Gamma \cup \{L\} \vdash T$ does not certify $\Gamma \vdash T$ unless the dependency on L is discharged or a separate certified $\Gamma \vdash L$ is provided. "Verifier accepted" must therefore specify the frozen judgment and assumption check, not merely validate local syntactic steps.

A permissible information flow is

$$\text{Imagine} \longrightarrow \text{Navigate} \longrightarrow \text{Propose subgoal} \longrightarrow \text{Prove} \longrightarrow \text{Verify} \longrightarrow \mathsf{BANK}.$$

4 MAPQUEST navigation inside a branch

For each σ, MAPQUEST may estimate backward cones from T, missing premises in target-producing inference frames, residual search work $\widehat{h}(T \mid \widetilde{B}_\sigma)$, enabled rules, Hilbert-address neighborhoods, candidate intermediate formulas, and alternative legal routes. Because $\widetilde{B}_\sigma$ may contain unproved hypotheses, every estimate is tagged *conditional on* H_σ. These maps may prioritize genuine subgoals but cannot be used as certificates of theoremhood.

5 High-leverage imagined lemmas and option gain

Let $\widehat{d}(T \mid B)$ be a declared heuristic residual estimate, not necessarily an admissible lower bound. For candidate L, define the *estimated* leverage

$$\widehat{\Delta}_T(L \mid B) = \widehat{d}(T \mid B) - \widehat{d}(T \mid B \cup \{L\}).$$

A positive value proposes L as a promising subtarget; it does not establish a genuine shortest-path decrease, or even that L is provable. A proposed threshold $\theta > 0$ may select $\widehat{\Delta}_T \geq \theta$.

An unbounded option-gain cardinality $|\operatorname{Enabled}(B \cup \{L\}) \setminus \operatorname{Enabled}(B)|$ may be infinite. Fix a finite, recorded window $\mathcal{C}_b$ of labeled candidate *rule applications* (not only their conclusion formulas); each application q has a finite premise set $\operatorname{Prem}(q)$. Define

$$\begin{aligned}
\operatorname{Enabled}_b^{\mathrm{cert}}(B) &= \{q \in \mathcal{C}_b : \operatorname{Prem}(q) \subseteq B\},\\
\operatorname{Enabled}_b^{\mathrm{hyp}}(B, H) &= \{q \in \mathcal{C}_b : \operatorname{Prem}(q) \subseteq B \cup H\},\\
\operatorname{Gain}_b(L \mid B) &= |\operatorname{Enabled}_b^{\mathrm{hyp}}(B, \{L\}) \setminus \operatorname{Enabled}_b^{\mathrm{cert}}(B)|.
\end{aligned}$$

The superscript hyp records conditional availability, never permission to write a purported rule consequence to BANK. Repeated rule applications with identical conclusions remain distinguishable. A finite-window heuristic can be of the form

$$\operatorname{Score}_b(L) = \alpha\widehat{\Delta}_T(L \mid B) + \beta\operatorname{Novelty}_b(L) + \gamma\operatorname{Gain}_b(L \mid B) - \delta\widehat{\operatorname{Cost}}_b(L),$$

with documented scales, sign conventions, and separately charged search/verification costs. This is a proposal-selection rule, not an optimality theorem.

6 Strategic sacrifice, recursion, and diagnostics

A search controller may accept a temporary increase in estimated residual work in exchange for a large set of future legal subgoals. The schematic objective

$$\operatorname{Sac}_b(a) = \widehat{\operatorname{FutureGain}}_b(a) - \lambda\widehat{\operatorname{ImmediateLoss}}_b(a)$$

ranks such moves without changing the verifier or licensing unsound inference. Repeated "what if L_1? what if L_2?" questions form a budgeted tree or DAG. The branch is evaluated conditionally, and any candidate returned to the certified environment is a subgoal, never a BANK fact.

When a declared stagnation diagnostic fires, a *Dreamer depth charge* may: checkpoint certified state; raise speculative depth within a budget; sample unusual branches, non-greedy priorities, and low-frequency address regions; measure conditional leverage and option gain; nominate concrete proof tasks; verify independently before any BANK deposit; and resume ordinary search. The diagnostic does not prove that stagnation is escapable or that the new schedule is cheaper.

7 Role and architecture discipline

In the AMLD+0 architecture, the six roles are 0 (Player-0), P (Prover), R (Refuter), D (Dependence), I (Independence), and C (Contradiction). A Dreamer or Analyst may be implemented as a *nonvoting internal tactic/module*, or in a separately specified architecture; neither designation replaces a settlement role. Speculative work may be delegated legally by the controller, with all costs charged; settled/refuted/dependence/independence/contradiction outcomes remain distinct.

8 Minimal operational algorithm

```
IMAGINE(B, T, depth, budget):
    queue := [(empty_address, empty_hypotheses, budget)]
    node_count := 1   # explicit finite cap N_max
    discoveries := []
    while queue is not empty and node_count <= N_max:
        (address, H, reserve) := take_next(queue)
        analyze MAPQUEST(B union H, T, status="speculative")
        if depth(address) < depth and reserve permits:
            for L in selected_candidate_hypotheses(address, H):
                child := (address append L, H union {L}, remaining_reserve)
                if node_count < N_max and charge_ok(child):
                    append child to queue       # tagged; no BANK write
                    node_count := node_count + 1
                if analysis makes L promising:
                    discoveries.append(proof_subgoal(Gamma |- L))
                    # goal status is UNVERIFIED; any branch proof is conditional
    return discoveries                          # unverified suggestions only

FOR EACH returned proof_subgoal:
    run authorized proof search (fully charged)
    require verifier receipt for the frozen judgment
    write to BANK only on accepted, assumption-safe receipt
```

The bounded queue is illustrative pseudocode and has not been executed as a certified general-purpose ATP. The test `charge_ok` must include the cost of generating and evaluating even discarded candidates, and an actual implementation must guarantee that scoring and candidate generation terminate or are interrupted at a charged cap.

9 Conclusion and limitations

Recursive imagination is *branch-preserving counterfactual search plus navigation plus hypothesis-safe promotion*. Theorem 3.1 formalizes a conditional safety invariant. It does not establish a successful escape policy, search completeness, verified runtime improvement, or a useful global metric. An executable implementation would need independent validation of its branch generator, assumption-dependency export, verifier trust boundary, and charged budget semantics. The pseudocode alone does not establish an executed or certified general-purpose ATP.